

# PROJECT REPORT

## Automated Network Request Management In ServiceNow

|                     |                                                    |
|---------------------|----------------------------------------------------|
| <b>Date</b>         | 20 September 2026                                  |
| <b>Team ID</b>      | Shaik Irshad - 238W1A4256                          |
| <b>Project Name</b> | Automated Network Request Management In ServiceNow |

## **1. Introduction**

### **1.1 Project Overview**

Automated Network Request Management In ServiceNow delivers a standardized, low-code ServiceNow workflow that lets an employee raise a network-related service request through the Service Portal and have ServiceNow automatically handle backend record creation, notification, approval routing, and status tracking end to end. It is built entirely using native ServiceNow capabilities — a Service Portal catalog item, Catalog Variables, a Catalog UI Policy, a User Information Variable Set, a custom scoped-application table, and a Flow Designer flow — with no custom Script Includes, Business Rules, or external integrations.

### **1.2 Purpose**

Conventional network-service requests handled through email, chat, or walk-in conversations lead to inconsistent information, manual data entry, difficult status tracking, manual approval coordination, and delayed notifications. The purpose of this project is to replace that fragmented process with a single, standardized ServiceNow workflow that validates and auto-populates request information, creates a traceable backend record automatically, and manages approval and status updates without manual intervention — improving consistency, visibility, and turnaround time for both the requester and the IT team.

## 2. Ideation Phase

### 2.1 Problem Statement

Two problem statements were defined using the “I am / I'm trying to / But / Because / Which makes me feel” format, capturing the requester's situation, goal, obstacle, root cause and resulting emotion.



*PS-1: Manual, Inconsistent Request Submission*



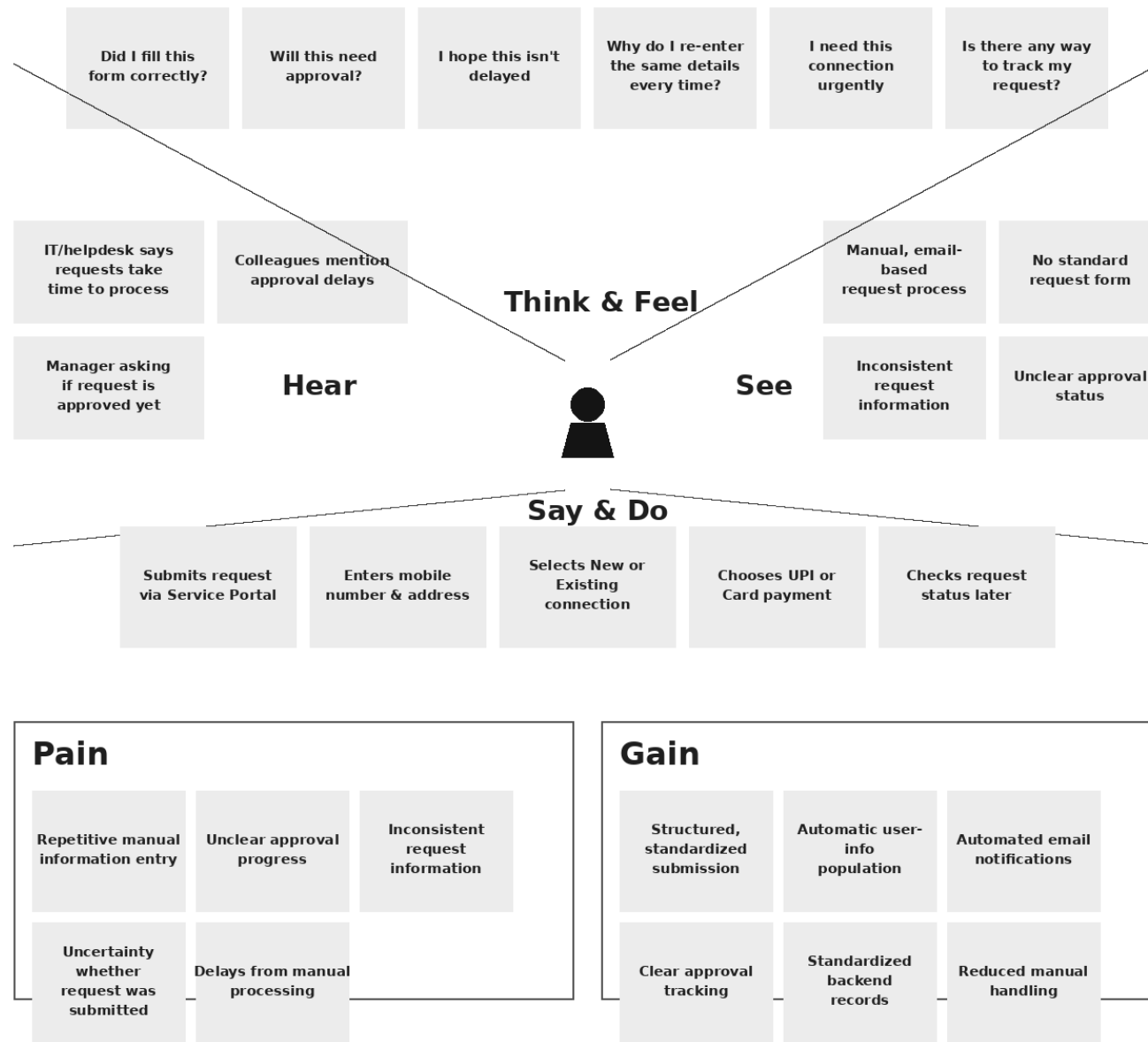
*PS-2: Lack of Approval Visibility*

Problem: Employees who need network-related services face a fragmented, largely manual request process where information must be collected consistently, approvals coordinated separately, and request progress is difficult to track.

Solution: A ServiceNow Service Portal-based Network Request catalog item, connected to a custom backend table and a Flow Designer automation, that handles record creation, email notification, approval, and status updates end to end.

## 2.2 Empathy Map Canvas

The empathy map below is built around the Requester persona — an employee submitting a network request — and is used for ideation purposes rather than as the output of a formal user study.



## 2.3 Brainstorming

Ideas were generated freely, then grouped by theme, and finally prioritized on importance versus feasibility to decide the project's actual scope.

### Step 2 – Brainstormed Ideas Grouped by Theme

#### Request Form & UI

Standardized  
Network  
Request catalog  
item

Mandatory-field  
validation

Dynamic  
show/hide of  
Existing ID  
field

Choice fields  
for  
connection &  
payment type

#### User & Backend Data

Auto-populate  
requester  
info (email,  
name, phone)

Custom backend  
table  
for requests

Auto-generated  
request  
/ database  
number

Approval Status  
field  
on backend  
record

#### Automation & Approval

Flow Designer  
instead  
of custom  
scripting

Service Catalog  
trigger  
on submission

Built-in Ask  
for  
Approval action

Approved /  
Rejected  
conditional  
branch

#### Notification & Tracking

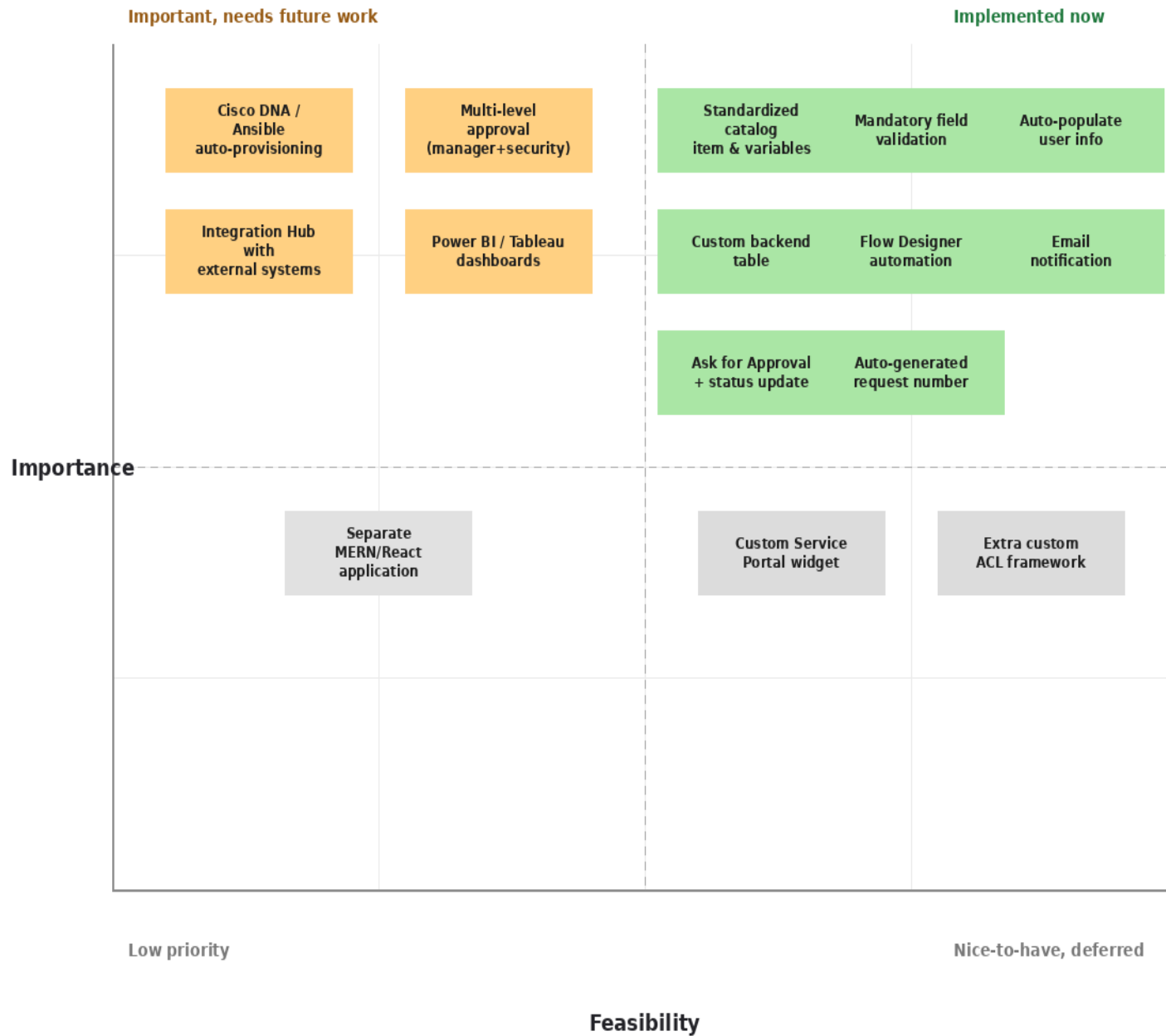
Automatic email  
on  
request  
creation

Approval  
related list  
on request  
record

Flow execution  
logs for  
troubleshooting

End-to-end  
status  
traceability

*Brainstormed ideas grouped by theme*



*Idea prioritization: implemented now (green) vs. future enhancement (orange) vs. out of scope (grey)*

## 3. Requirement Analysis

### 3.1 Customer Journey Map

The map traces the Requester's journey from first realizing a network connection is needed through to seeing the final Approved/Rejected outcome, across the phases Entice → Enter → Engage → Exit → Extend.

SERVICENOW

#### Network Request Journey

SCENARIO  
Discovering, submitting, and tracking a network service request end to end

|                                                                                                                                                               | Entice                                  |                                              | Enter                                        |                                              | Engage                                          |                                             |                                         | Exit                                    |                                             | Extend                                     |                                              |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|----------------------------------------------|----------------------------------------------|----------------------------------------------|-------------------------------------------------|---------------------------------------------|-----------------------------------------|-----------------------------------------|---------------------------------------------|--------------------------------------------|----------------------------------------------|
|                                                                                                                                                               | How does the requester...?              |                                              | What happens as they...?                     |                                              | In the core moments...                          |                                             |                                         | What happens as the...?                 |                                             | What happens after the...?                 |                                              |
| <br><b>Steps</b><br>What does the requester typically experience?            | Realizes a network connection is needed | Learns about the IT Service Portal           | Logs into the Service Portal                 | Finds the Network Request catalog item       | Fills in requester & connection details         | Enters Existing ID if applicable            | Enters payment info & submits           | Sees on-screen submission confirmation  | Receives email with request number          | Approver reviews & decides                 | Requester checks final status                |
| <br><b>Interactions</b><br>Who/where/what do they interact with?             | Self; own workspace/desk                | IT colleague; team chat                      | Service Portal login (SSO)                   | Service Portal; Network category             | Network Request form; Catalog Variables         | Existing ID field (Catalog UI Policy)       | Payment fields; Submit button           | Confirmation screen / request number    | Email inbox; Flow-sent notification         | sysapproval_app rover record               | Approval Related List; Approval Status field |
| <br><b>Goals &amp; Motivations</b><br>"Help me..." / "Help me avoid..."      | Help me know what to request            | Help me find the right place to ask          | Help me get in without hassle                | Help me find the catalog item fast           | Help me avoid re-typing my info                 | Help me avoid mistakes if already connected | Help me not miss a required field       | Help me know it really went through     | Help me keep a reference number             | Help me get a fair, timely decision        | Help me know exactly where I stand           |
| <br><b>Positive Moments</b><br>What feels easy, fast, or reassuring?        | Clear that IT needs can become a ticket | Colleague points them straight to the portal | Single sign-on makes login painless          | "Network" category is easy to browse         | Requester info auto-populates itself            | Field only appears when actually needed     | Validation catches mistakes early       | Instant confirmation feels reassuring   | Email arrives with a clear request number   | Routed automatically — no manual chasing   | Status visible in one place, no guesswork    |
| <br><b>Negative Moments</b><br>What feels confusing, slow, or frustrating? | Not obvious IT can help with this       | Portal existence isn't widely known          | Portal feels unfamiliar for infrequent users | Many categories can confuse first-time users | Connection-specific details still typed by hand | "Existing ID" unclear without guidance      | Submission blocked if a field is missed | No visibility yet into who is approving | Notification could get lost in a busy inbox | No estimate of how long approval will take | No nudge if a request sits pending too long  |
| <br><b>Areas of Opportunity</b><br>How might we make each step better?     | Promote this in IT onboarding / FAQs    | Add a portal shortcut on intranet pages      | Offer a quick-start guide for new users      | Add search keywords to speed up discovery    | Add contextual help next to unclear fields      | Add a tooltip on the Existing ID field      | Show validation messages inline, sooner | Show the assigned approver on screen    | Add a reminder if approval is pending       | Track average approval turnaround time     | Send a proactive status-changed alert        |

### 3.2 Solution Requirement

#### Functional Requirements:

| FR No. | Functional Requirement (Epic)             | Sub Requirement (Story / Sub-Task)                                                                                                                                                           |
|--------|-------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FR-1   | <b>Network Request Submission</b>         | Submit a Network Request through the ServiceNow Service Portal<br>Enter requester, connection, payment and address details<br>Mandatory-field validation before submission                   |
| FR-2   | <b>Dynamic Form Behaviour</b>             | Show the Existing ID field only when Type of Connection = Existing (Catalog UI Policy)                                                                                                       |
| FR-3   | <b>Requester Info Auto-Population</b>     | Auto-populate Email ID, User Name and Phone Number from “Opened on behalf of” (Variable Set)                                                                                                 |
| FR-4   | <b>Backend Record Creation</b>            | Create a record in the custom Database Tables table on submission<br>Map catalog variables to backend fields<br>Auto-generate the Database Number<br>Set initial Approval Status = Requested |
| FR-5   | <b>Email Notification</b>                 | Send an email to the requester with request info and the generated request number                                                                                                            |
| FR-6   | <b>Approval Workflow</b>                  | Create an approval request (Ask for Approval)<br>Evaluate the decision and update Approval Status to Approved/Rejected                                                                       |
| FR-7   | <b>Status Tracking &amp; Traceability</b> | Display related approval records via the Approval Request relationship<br>Provide end-to-end visibility from submission to final status                                                      |

#### Non-Functional Requirements:

| FR No. | Non-Functional Requirement | Description                                                                                                                                                           |
|--------|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| NFR-1  | <b>Usability</b>           | A single standardized Service Portal form with mandatory-field validation and dynamic show/hide of the Existing ID field reduces confusion and repeated manual entry. |
| NFR-2  | <b>Security</b>            | Access to the custom table is governed by ServiceNow's default, auto-generated ACLs; all traffic runs over the platform's standard HTTPS/TLS.                         |
| NFR-3  | <b>Reliability</b>         | Flow Designer executes the same fixed sequence on every submission; verified using Flow Execution Details, email logs and record checks.                              |
| NFR-4  | <b>Performance</b>         | Flow Designer runs asynchronously, so the requester's Service Portal experience is not blocked while backend processing completes.                                    |
| NFR-5  | <b>Availability</b>        | Runs on ServiceNow's SLA-backed, multi-instance cloud hosting; availability is platform-managed.                                                                      |
| NFR-6  | <b>Scalability</b>         | As a scoped application on the Now Platform, the solution inherits native multi-instance cloud scalability with no custom scaling logic.                              |



### 3.3 Data Flow Diagram

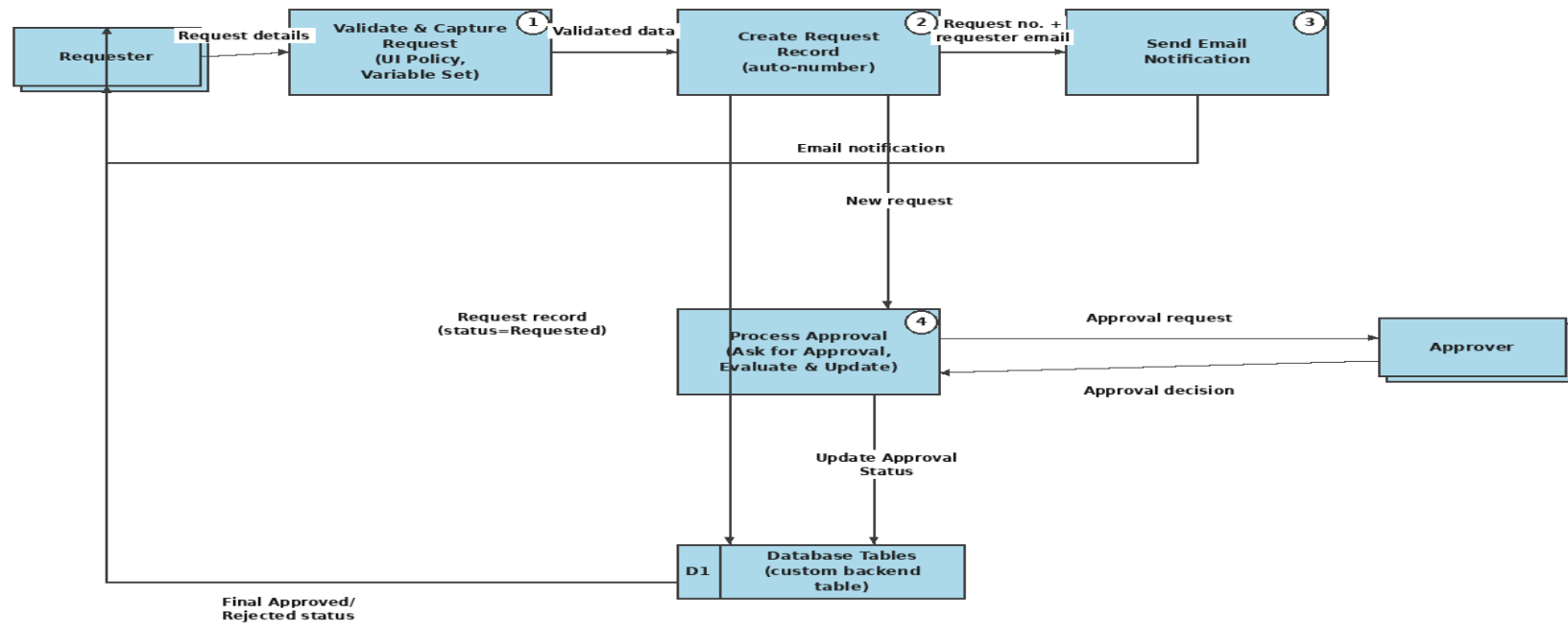
Level-0 (context) and Level-1 (detailed) data flow diagrams describe how request data moves through the system.

**Level-0 DFD (Context Diagram)**



*Level-0 DFD (Context Diagram)*

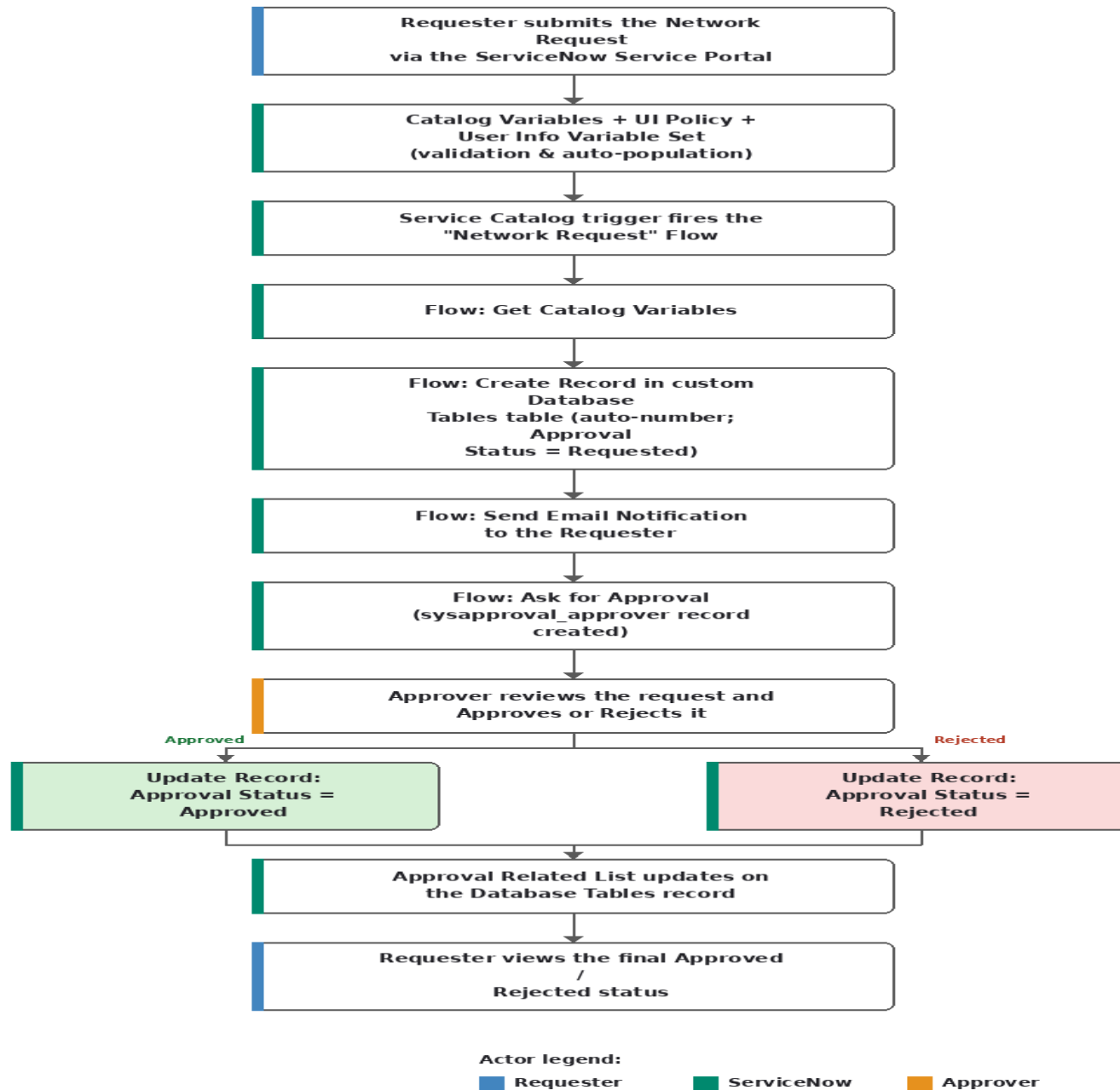
**Level-1 DFD — Network Request Process**



*Level-1 DFD — Network Request Process*

### 3.4 Technology Stack

#### Network Request - Technical Flow



*Technical flow of the Network Request implementation*

| S.No | Component                     | Description                                                        | Technology                                                          |
|------|-------------------------------|--------------------------------------------------------------------|---------------------------------------------------------------------|
| 1    | <b>User Interface</b>         | How the requester interacts with the application                   | ServiceNow Service Portal - "Network Request" Catalog Item          |
| 2    | <b>Application Logic-1</b>    | Captures and validates/auto-populates request details              | Catalog Variables, Catalog UI Policy, User Information Variable Set |
| 3    | <b>Application Logic-2</b>    | Orchestrates record creation, notification and approval end to end | ServiceNow Flow Designer ("Network Request" flow)                   |
| 4    | <b>Application Logic-3</b>    | Routes the request for approval and evaluates the outcome          | ServiceNow native Approval Engine (sysapproval_approver)            |
| 5    | <b>Database</b>               | Stores submitted network-request data                              | Custom "Database Tables" table in the scoped application            |
| 6    | <b>Cloud Database</b>         | Underlying data store                                              | ServiceNow multi-instance cloud database (no separate external DB)  |
| 7    | <b>File Storage</b>           | Attachment storage                                                 | Not implemented — not required by the current form                  |
| 8    | <b>External API-1/2</b>       | External systems consumed                                          | Not implemented in this project (see Future Scope)                  |
| 9    | <b>Machine Learning Model</b> | ML usage                                                           | Not applicable / not implemented                                    |
| 10   | <b>Infrastructure</b>         | Where the app is deployed                                          | ServiceNow multi-tenant cloud instance, fully platform-managed      |

## 4. Project Design

### 4.1 Problem Solution Fit

The Problem-Solution Fit canvas checks that the realized ServiceNow solution actually addresses the requester's real behaviour, constraints and triggers.

#### Problem - Solution Fit Canvas — Network Request (ServiceNow)

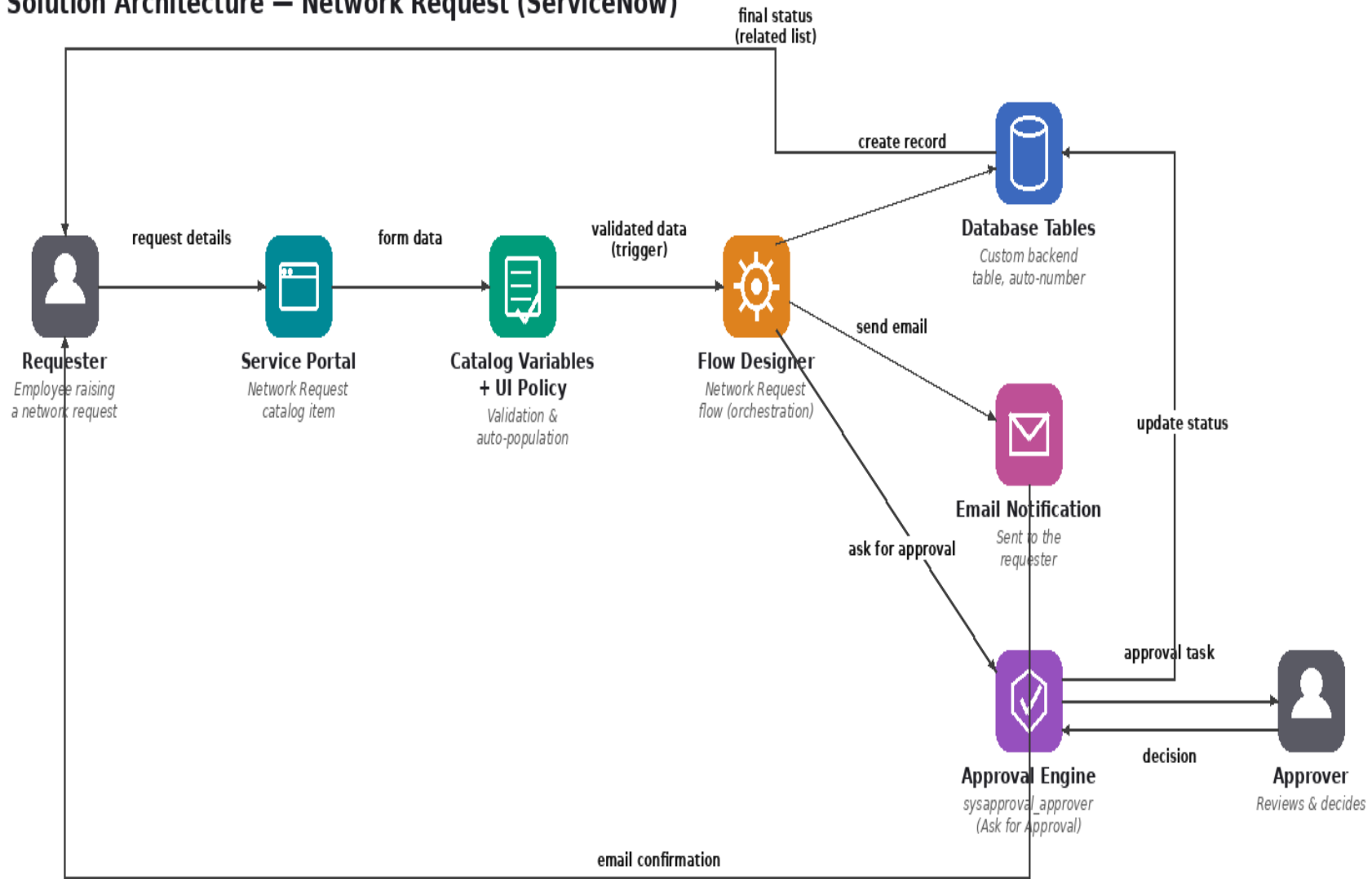
|                           |                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                        |                             |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|
| Define CS, fit into CC    | <b>1. CUSTOMER SEGMENT(S)</b> <span>CS</span><br>Employees across the organization who occasionally need network-related IT services — e.g. a new connection at a new desk/office, or a change to an existing connection.                                                                  | <b>6. CUSTOMER CONSTRAINTS</b> <span>CC</span><br>No direct access to network infrastructure or ticketing expertise; must go through official IT channels; limited time to chase status manually; dependent on IT/network team availability and the approval hierarchy; cannot self-provision.                                                                                                                                                                                                                    | <b>5. AVAILABLE SOLUTIONS</b> <span>AS</span><br>Before this project: emailing IT support directly, an unstructured helpdesk ticket, or a walk-in request. Pros: familiar. Cons: inconsistent information, approvals coordinated manually over email/chat, no automatic tracking or status visibility. | Explore AS, differentiate   |
|                           | <b>2. JOBS-TO-BE-DONE / PROBLEMS</b> <span>J&amp;P</span><br>Get a new or existing network connection set up correctly and on time; submit request details once without re-entering known information; know whether and when the request was approved; avoid delays from miscommunication. | <b>9. PROBLEM ROOT CAUSE</b> <span>RC</span><br>No standardized, self-service system existed for network requests. Intake, approval and status tracking were handled through disconnected manual channels (email/chat/walk-in) instead of one structured workflow with a single source of truth.                                                                                                                                                                                                                  | <b>7. BEHAVIOUR</b> <span>BE</span><br>Directly related: find IT/network contact, email or message them the request, wait for a reply, follow up or escalate to a manager if delayed. Indirectly related: ask a colleague who raised a similar request, or search the intranet/FAQ.                    |                             |
| Identify strong TR & EM   | <b>3. TRIGGERS</b> <span>TR</span><br>Moving to a new desk/office; onboarding into a team needing network access; an existing connection stops working; a manager/colleague suggests raising an IT request.                                                                                | <b>10. YOUR SOLUTION</b> <span>SL</span><br>Built on the organization's existing ServiceNow Service Portal: a standardized "Network Request" catalog item with mandatory-field validation and auto-populated requester details, a Flow Designer automation that creates the backend record, sends an email, and routes the request through ServiceNow's native approval engine, then updates the final Approved/Rejected status — one channel, one visible source of truth, no manual re-entry or status chasing. | <b>8. CHANNELS of BEHAVIOUR — 8.1 ONLINE</b> <span>CH</span><br>Company email; team chat (e.g. Teams/Slack); intranet/FAQ search; the ServiceNow Service Portal itself.                                                                                                                                | Extract online & offline CH |
|                           | <b>4. EMOTIONS: BEFORE / AFTER</b> <span>EM</span><br>Before: uncertain, worried about doing it "the wrong way". After (old process): anxious, left waiting with no visibility. After (this solution): reassured and in control — confirmed instantly, status always visible.              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>8. CHANNELS of BEHAVIOUR — 8.2 OFFLINE</b> <span>CH</span><br>Walking up to the IT help desk in person; asking a colleague directly; phoning IT support.                                                                                                                                            |                             |
| Focus on J&P, tap into BE |                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                        | Understand RC               |

## 4.2 Proposed Solution

| S.No. | Parameter                                    | Description                                                                                                                                                                                                                                                                                       |
|-------|----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1     | <b>Problem Statement</b>                     | <ul style="list-style-type: none"><li>Employees face a fragmented, manual network-request process with inconsistent information, manual approval coordination, and poor status visibility.</li></ul>                                                                                              |
| 2     | <b>Idea / Solution</b>                       | <ul style="list-style-type: none"><li>Standardized Service Portal catalog item with validation and dynamic fields; auto-populated requester info; Flow Designer creates the backend record, sends email, and manages approval; full traceability via the Approval Request relationship.</li></ul> |
| 3     | <b>Novelty / Uniqueness</b>                  | <ul style="list-style-type: none"><li>A fully configuration-first, low-code build — no custom Script Includes or Business Rules — using only native ServiceNow capabilities end to end.</li></ul>                                                                                                 |
| 4     | <b>Social Impact / Customer Satisfaction</b> | <ul style="list-style-type: none"><li>Removes repetitive manual entry, gives automatic confirmation, and provides clear approval visibility, reducing follow-up queries and turnaround time.</li></ul>                                                                                            |
| 5     | <b>Business Model</b>                        | <ul style="list-style-type: none"><li>An internal IT-efficiency solution: value comes from reduced IT support hours and fewer errors, not direct revenue; could be packaged as a reusable accelerator as a future direction.</li></ul>                                                            |
| 6     | <b>Scalability</b>                           | <ul style="list-style-type: none"><li>Runs natively on ServiceNow's multi-instance cloud; the same pattern can be extended to other IT service requests by cloning the existing components.</li></ul>                                                                                             |

### 4.3 Solution Architecture

#### Solution Architecture – Network Request (ServiceNow)



## 5. Project Planning & Scheduling

### 5.1 Project Planning

#### Product Backlog:

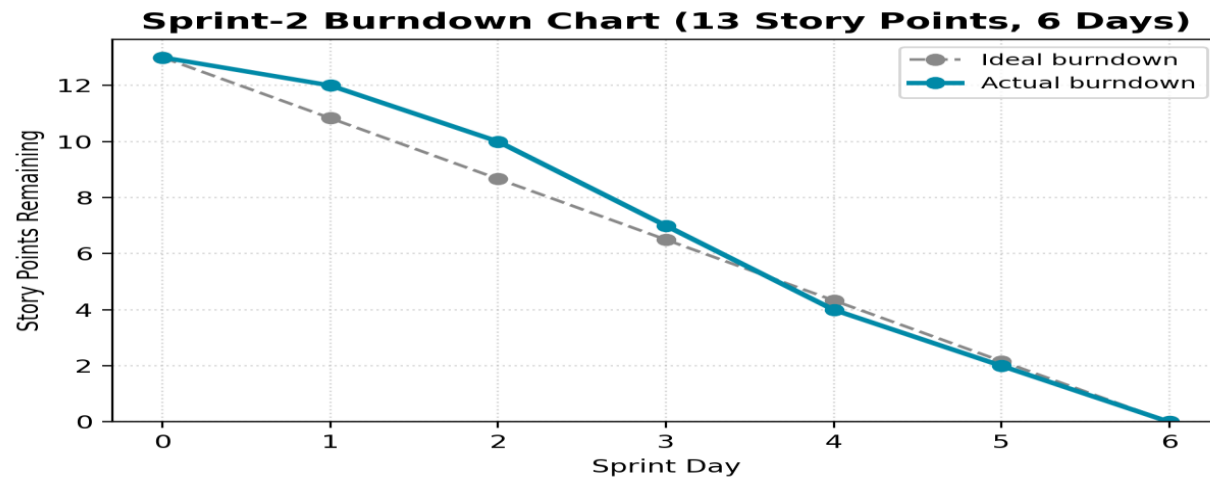
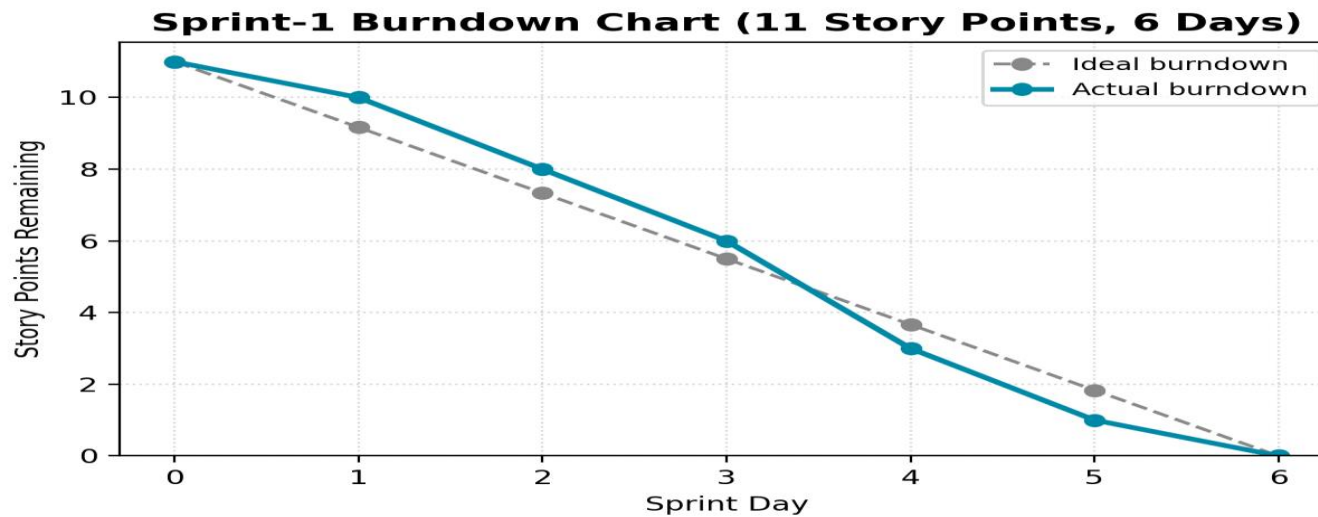
| Sprint   | Epic                           | Story No.  | User Story                                                                                 | Points | Priority |
|----------|--------------------------------|------------|--------------------------------------------------------------------------------------------|--------|----------|
| Sprint-1 | Network Request Submission     | USN-1      | As a requester, I can open the Network Request catalog item and enter my request details.  | 2      | High     |
| Sprint-1 | Network Request Submission     | USN-2      | As a requester, I cannot submit the request if a mandatory field is left blank.            | 1      | High     |
| Sprint-1 | Network Request Submission     | USN-3      | As a requester, I can select whether my connection is New or Existing.                     | 1      | High     |
| Sprint-1 | Dynamic Form Behaviour         | USN-4      | As a requester, I only see the Existing ID field when I select Existing connection.        | 2      | Medium   |
| Sprint-1 | Requester Info Auto-Population | USN-5      | As a requester, my email, name and phone number are auto-populated.                        | 2      | Medium   |
| Sprint-1 | Backend Configuration          | USN-11     | As an admin, I can see all submitted requests as structured records in one table.          | 3      | Medium   |
| Sprint-2 | Email Notification             | USN-6      | As a requester, I receive an email once my request has been created.                       | 2      | High     |
| Sprint-2 | Status Tracking & Traceability | USN-7      | As a requester, I can check whether my request is pending, Approved, or Rejected.          | 2      | High     |
| Sprint-2 | Approval Workflow              | USN-8/9/10 | As an approver, I receive, review, and decide on requests, which updates the final status. | 4      | High     |
| Sprint-2 | Backend Configuration          | USN-12     | As an admin, I can trace an approval back to its request record.                           | 2      | Medium   |
| Sprint-2 | Security & Access              | USN-13     | As an admin, the table is secured using default ACLs without a custom framework.           | 1      | Low      |
| Sprint-2 | Troubleshooting                | USN-14     | As an admin, I can diagnose why a submission did not trigger automation.                   | 2      | Medium   |

#### Sprint Schedule:

| Sprint   | Total Points | Duration | Start Date  | End Date (Planned) | Points Completed | Release Date (Actual) |
|----------|--------------|----------|-------------|--------------------|------------------|-----------------------|
| Sprint-1 | 11           | 6 Days   | 01 Aug 2026 | 06 Aug 2026        | 11               | 06 Aug 2026           |
| Sprint-2 | 13           | 6 Days   | 08 Aug 2026 | 13 Aug 2026        | 13               | 13 Aug 2026           |

Velocity: with a 6-day sprint duration and 24 combined story points across two sprints,  $AV = \text{Velocity} / \text{Sprint Duration} = 24 / 12 = 2$  story points/day.

$$AV = \frac{\text{Velocity}}{\text{Sprint Duration}} = \frac{24}{12} = 2 \text{ story points/day}$$



*Sprint-1 and Sprint-2 burndown charts — both sprints completed on schedule*



## 6. Functional and Performance Testing

The table below lists the functional test cases executed via actual Service Portal submissions, covering validation, dynamic behaviour, backend record creation, notification, and the approval workflow.

| Test Case ID | Description                                                | Expected Result                                                                      | Actual Result                                                                                                               | Status           |
|--------------|------------------------------------------------------------|--------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|------------------|
| TC-01        | Submit with a mandatory field left blank                   | Submission is blocked with a validation message                                      | Blocked as expected                                                                                                         | Pass             |
| TC-02        | Select 'Existing' connection type                          | "Enter your Existing ID" field becomes visible                                       | Field appeared                                                                                                              | Pass             |
| TC-03        | Select 'New' connection type                               | Existing ID field stays hidden                                                       | Field hidden                                                                                                                | Pass             |
| TC-04        | Set 'Opened on behalf of'                                  | Email, User Name and Phone Number auto-populate                                      | Auto-populated correctly                                                                                                    | Pass             |
| TC-05        | Submit a fully valid request                               | New record created in Database Tables with a unique auto-number                      | Record created with number                                                                                                  | Pass             |
| TC-06        | Check newly created record                                 | Approval Status = Requested                                                          | Status = Requested                                                                                                          | Pass             |
| TC-07        | Submit a valid request                                     | Requester receives an email with the request number                                  | Email received                                                                                                              | Pass             |
| TC-08        | Submit a valid request                                     | sysapproval_approver record created and linked via the Approval Request relationship | Approval record linked                                                                                                      | Pass             |
| TC-09        | Approver approves the request                              | Approval Status updates to Approved                                                  | Status updated                                                                                                              | Pass             |
| TC-10        | Approver rejects the request                               | Approval Status updates to Rejected                                                  | Status updated                                                                                                              | Pass             |
| TC-11        | Submit via Service Portal (initial catalog item/flow link) | Flow executes automatically on submission                                            | Initially did not trigger; passed after correcting the catalog item's Flow/Process Engine association (see Troubleshooting) | Pass (after fix) |
| TC-12        | A different user views another requester's record          | Access governed by ServiceNow's default ACLs                                         | Access controlled as expected                                                                                               | Pass             |

### 6.1 Performance Testing

As a solution built entirely on ServiceNow's managed SaaS platform, performance testing was scoped to what the project team could directly observe rather than infrastructure load-testing:

- Flow Designer executes asynchronously, so the Service Portal submission returns to the requester immediately without waiting on record creation, email, or approval steps.
- During testing, the time from submission to backend record creation and email dispatch was consistently within a few seconds.
- Approval creation and status-update steps were verified through Flow Execution Details with no noticeable delay.
- No formal load or stress testing (e.g. many concurrent submissions) was conducted, as scaling and infrastructure performance are managed by the ServiceNow platform itself and were outside this project's scope.

# 7. Results

## 7.1 Output Screenshots

This section should contain screenshots captured directly from your ServiceNow instance — they cannot be generated on your behalf without access to your live instance. Please insert screenshots of the following before final submission:

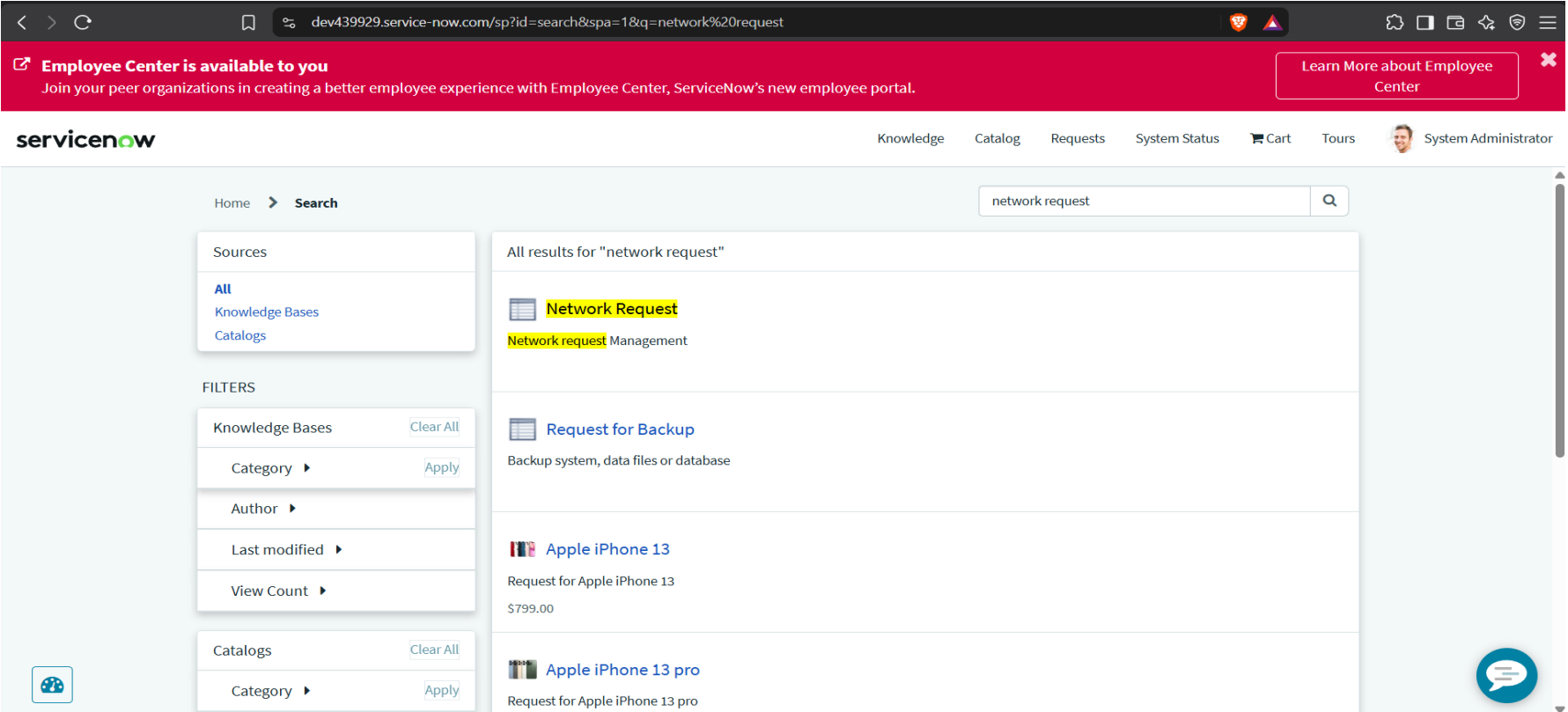


Fig-1:The Network Request catalog item as seen in the Service Portal

| Catalog Items                                                                                                   |                 |                            |        |        |                 |                          |        |         |                     |
|-----------------------------------------------------------------------------------------------------------------|-----------------|----------------------------|--------|--------|-----------------|--------------------------|--------|---------|---------------------|
| Application scope: Automated Net...<br>Update set: System Administrator [...]                                   |                 |                            |        |        |                 |                          |        |         |                     |
| All > Type != Bundle > Class != Order guide > Type != Package > Class != Content Item > Published item is empty |                 |                            |        |        |                 |                          |        |         |                     |
|                                                                                                                 | Name            | Short description          | Active | Roles  | Catalogs        | Category                 | Price  | Type    | Updated             |
| <input type="checkbox"/>                                                                                        | Search          | Search                     | Search | Search | Search          | Search                   | Search | !=Packa | Search              |
|                                                                                                                 | Network Request | Network request Management | true   |        | Service Catalog | Network Standard Changes | \$0.00 | Item    | 2026-09-18 20:04:03 |

Variables (7)

Variable Sets (1)

Catalog UI Policies (1)

Catalog Client Scripts

Available For

Not Available For

Categories (1)

Catalogs (1)

Catalog Data Lookup Definitions

Related Articles

Related Catalog Items

Assigned Topics

≡

🔍

Order

▼

Search

⚙️

📄

🔍

📋

—

Actions on selected rows...

▼

New

Catalog item = Network Request

|                          |              |                  |                        |         |
|--------------------------|--------------|------------------|------------------------|---------|
| <input type="checkbox"/> | <div>🔍</div> | Type             | Question               | Order ▲ |
|                          |              | Single Line Text | Requested For          | 100     |
|                          |              | Single Line Text | Mobile Number          | 200     |
|                          |              | Multiple Choice  | Type of Connection     | 300     |
|                          |              | Single Line Text | Enter your Existing ID | 400     |
| <input type="checkbox"/> | <div>ⓘ</div> | Single Line Text | Total Amount           | 500     |
|                          |              | Multiple Choice  | Mode of Payment        | 600     |
|                          |              | Single Line Text | Address                | 700     |

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1 to 7 of 7

⏴

⏵

Variables (7)

Variable Sets (1)

Catalog UI Policies (1)

Catalog Client Scripts

Available For

Not Available For

Categories (1)

Catalogs (1)

Catalog Data Lookup Definitions

Related Articles

Related Catalog Items

Assigned Topics

≡

🔍

Order

▼

Search

⚙️

📄

🔍

📄

—

Actions on selected rows...

▼

New

Catalog item = Network Request

|                          |   |                                          |              |            |                  |         |         |                     |         |
|--------------------------|---|------------------------------------------|--------------|------------|------------------|---------|---------|---------------------|---------|
| <input type="checkbox"/> | 🔍 | Short description                        | Variable set | Conditions | Reverse if false | On load | Inherit | Updated             | Order ▲ |
|                          |   | Show Existing ID for Existing Connection | (empty)      |            | true             | true    | false   | 2026-09-18 19:12:44 | 100     |

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⏩

1 to 1 of 1

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Catalog UI Policy

Show Existing ID for Existing Connection

🔗

?

⚙️

🔍

⋮

Update

Delete

↑

↓

Catalog Conditions

Add Filter Condition

Add OR Clause

type\_of\_connection

▼

is

▼

Existing

▼

AND

OR

✖

Applies on a Catalog Item view

☒

Applies on Catalog Tasks

☐

Applies on Requested Items

☐

Apply the catalog UI policy actions when the form is loaded or when the user changes values on the form

On load ☒

Reverse the effects of the catalog UI policy actions when the Conditions evaluate to false

Reverse if false ☒

Update

Delete

Related Links

[Run Point Scan](#)

[\[SN Utils\] Versions \(1\)](#)

|                                                      |   |                           |             |   |             |    |         |   |         |   |                             |   |     |
|------------------------------------------------------|---|---------------------------|-------------|---|-------------|----|---------|---|---------|---|-----------------------------|---|-----|
| ≡                                                    | 🔍 | Catalog UI Policy Actions | Order       | ▼ | Search      | ⚙️ | 📄       | 🔍 | 📄       | — | Actions on selected rows... | ▼ | New |
| UI policy = Show Existing ID for Existing Connection |   |                           |             |   |             |    |         |   |         |   |                             |   |     |
| <input type="checkbox"/>                             | 🔍 | Name                      | Read only   |   | Mandatory   |    | Visible |   | Order ▲ |   |                             |   |     |
|                                                      |   | enter_your_existing_id    | Leave alone |   | Leave alone |    | True    |   | 100     |   |                             |   |     |

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Catalog UI Policy Action  
enter\_your\_existing\_id

🔗

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⋮

Update

Delete

↑

UI policy actions specify exactly what actions to take on a specified field. The conditions specified in the UI policy determine when these actions are triggered. [More Info](#)

Catalog Item

Network Request

Variable name

enter\_your\_existing\_id

Order

100

Application

Automated Network Request Mana

ⓘ

Mandatory

Leave alone

▼

Visible

True

▼

Read only

Leave alone

▼

Value action

Leave alone

▼

Field message type

None

▼

Update

Delete

[\[N Utils\] Versions \(1\)](#)

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Variable  
Email ID

🔗

?

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Copy

Update

Delete

↑

↓

Application

Automated Network Request Mana

ⓘ

Type

Single Line Text

▼

Order

200

Variable set

User Information

🔍

ⓘ

Active

☒

Mandatory

☐

Read only

☐

Hidden

☐

Disable automatic slot fill  
based on user context

☐

Question

Annotation

Type Specifications

Default Value

Auto-populate

Permission

Availability

Dependent question

Opened on behalf of

🔍

ⓘ

Reference

User [sys\_user]

▼

Dot walk path

Email

▼

Copy

Update

Delete

Related Links  
[Run Point Scan](#)  
[\[SN Utils\] Versions \(2\)](#)

**Fig-3: The Catalog UI Policy and the Existing ID field's dynamic show/hide behaviour**

Database Number DT0001001



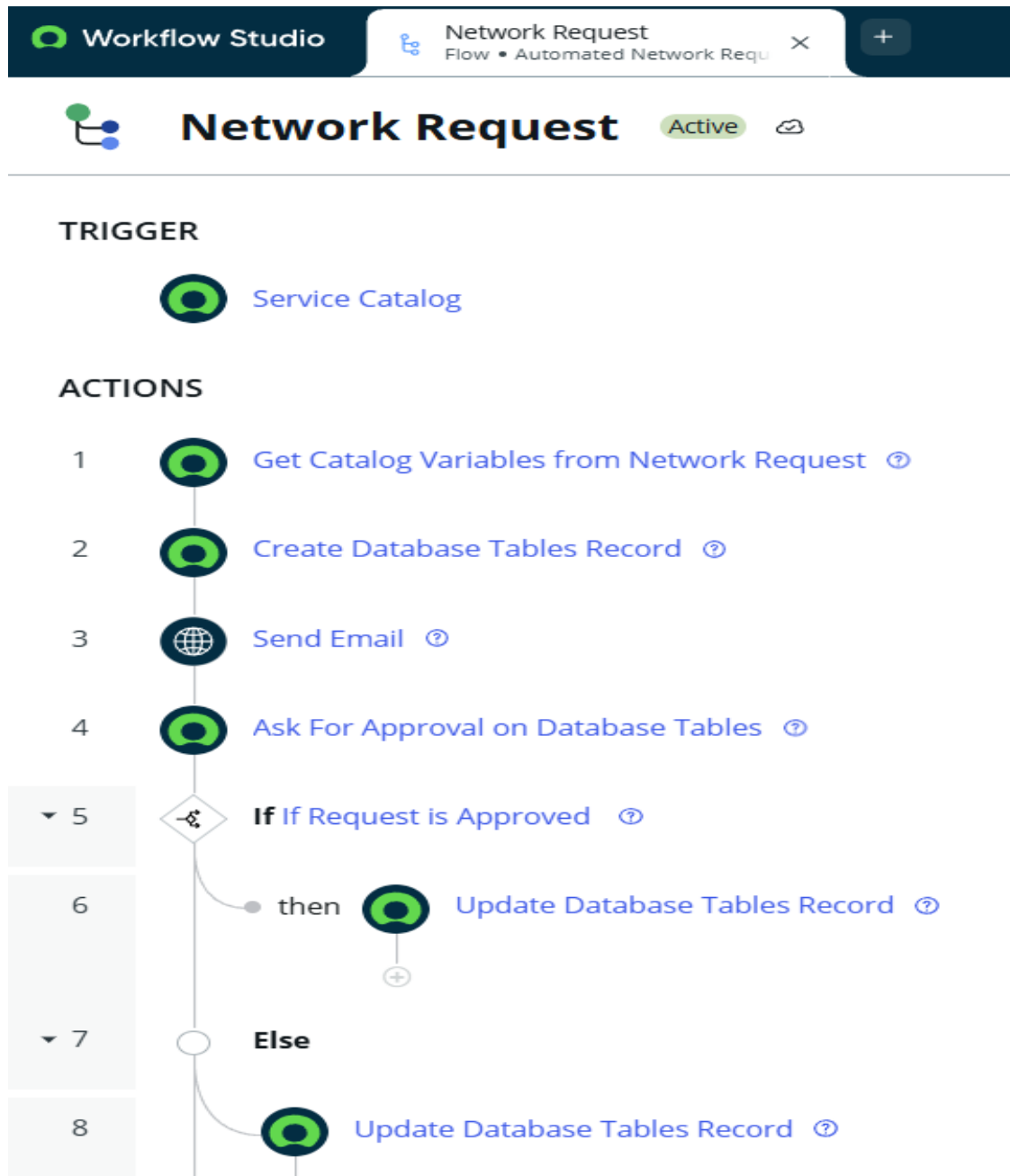


Fig-5: The Flow Designer canvas showing the complete “Network Request” flow

Approval Request

State

Search

⚙️

📄

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Actions on selected rows...

New

Approvals

| State     | Approver             | Comments | Approval for | Created             |
|-----------|----------------------|----------|--------------|---------------------|
| Requested | System Administrator |          | (empty)      | 2026-09-20 06:48:25 |

1 to 1 of 1

Workflow Studio

Network Request

Network Request

EXECUTION DETAILS

Network Request

Test Run - Waiting

Cancel flow

Open flow

Open context record

Show Action Details

|                 | State                                      | Start time     |                     |
|-----------------|--------------------------------------------|----------------|---------------------|
| FLOW STATISTICS | Run as: System Administrator               | Open flow logs | Waiting             |
| TRIGGER         |                                            |                | 2026-09-20 06:50:38 |
|                 |                                            |                | 83ms                |
| ACTIONS         |                                            |                |                     |
| 1               | Get Catalog Variables from Network Request | Core Action    | Completed           |
| 2               | Create Record                              | Core Action    | Completed           |
| 3               | Send Email                                 |                | Completed           |
| 4               | Ask For Approval                           | Core Action    | Waiting             |
| 5               | If If Request is Approved                  | Flow Logic     | Not Run             |
| 6               | Update Record                              | Core Action    | Not Run             |
| 7               | Else                                       | Flow Logic     | Not Run             |
| 8               | Update Record                              | Core Action    | Not Run             |

[SN Utils] Versions (0)

Approval Request

State

Search

⚙️

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Actions on selected rows...

New

Approvals

| State    | Approver             | Comments | Approval for | Created             |
|----------|----------------------|----------|--------------|---------------------|
| Approved | System Administrator |          | (empty)      | 2026-09-20 06:45:56 |





EXECUTION DETAILS

Network Request



Test Run - Completed

Open flow

Open context record

TRIGGER

 Catalog Item Requested

ACTIONS

1  Get Catalog Variables from Network Request

2  Create Record

3  Send Email

4  Ask For Approval

▼ 5  If If Request is Approved

6  Update Record

▼ 7  Else

8  Update Record

ERROR HANDLER

|             |                  |                     |      |
|-------------|------------------|---------------------|------|
| Core Action | Completed        | 2026-09-20 06:45:56 | 23ms |
| Core Action | Completed        | 2026-09-20 06:45:56 | 6ms  |
|             | Completed        | 2026-09-20 06:45:56 | 25ms |
| Core Action | Completed        | 2026-09-20 06:45:56 | 72ms |
| Flow Logic  | Evaluated - True | 2026-09-20 06:46:46 | 3ms  |
| Core Action | Completed        | 2026-09-20 06:46:46 | 3ms  |
| Flow Logic  | Not Run          |                     |      |
| Core Action | Not Run          |                     |      |

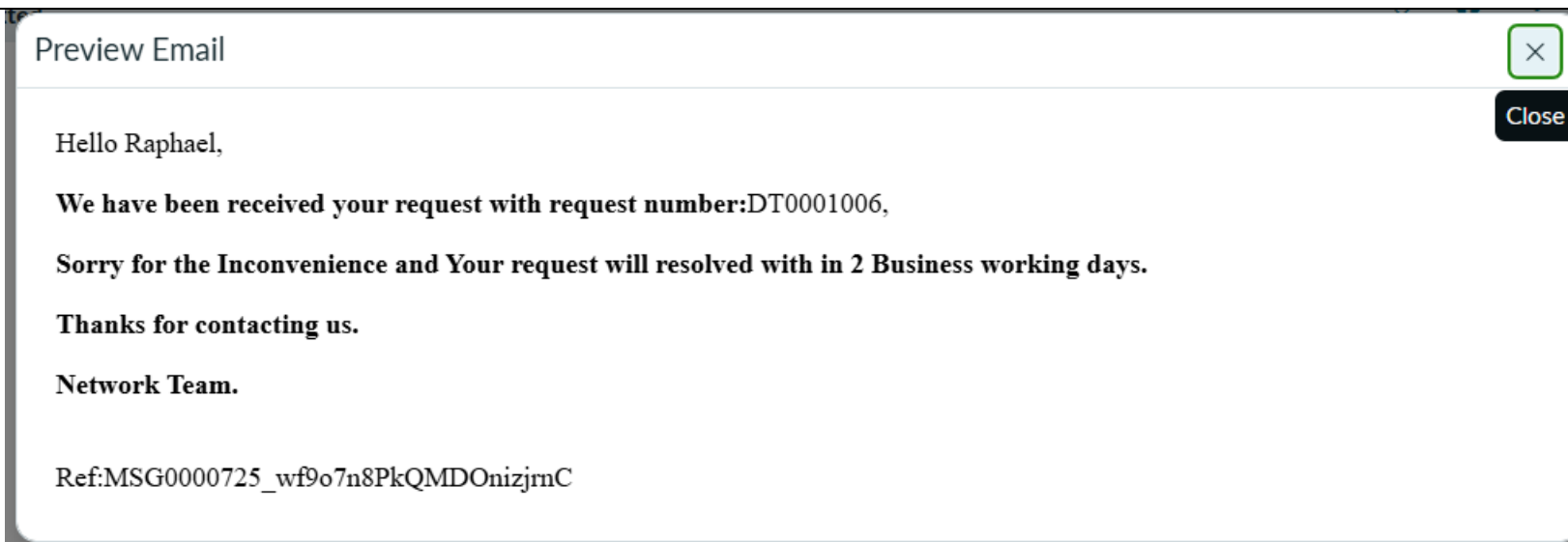


Fig-6: A submitted request record, the generated email notification, and the approval record

Workflow Studio

Network Request

EXECUTION DETAILS

Network Request

Test Run - Waiting

Cancel flow

Open flow

Open context record

Show Action Details

|                                              | State                        | Start time     |                     |
|----------------------------------------------|------------------------------|----------------|---------------------|
| FLOW STATISTICS                              | Run as: System Administrator | Open flow logs | Waiting             |
| TRIGGER                                      |                              |                |                     |
| Catalog Item Requested                       |                              |                |                     |
| ACTIONS                                      |                              |                |                     |
| 1 Get Catalog Variables from Network Request | Core Action                  | Completed      | 2026-09-20 06:50:38 |
| 2 Create Record                              | Core Action                  | Completed      | 2026-09-20 06:50:38 |
| 3 Send Email                                 |                              | Completed      | 2026-09-20 06:50:38 |
| 4 Ask For Approval                           | Core Action                  | Waiting        | 2026-09-20 06:50:38 |
| 5 If Request is Approved                     | Flow Logic                   | Not Run        |                     |
| 6 Update Record                              | Core Action                  | Not Run        |                     |
| 7 Else                                       | Flow Logic                   | Not Run        |                     |
| 8 Update Record                              | Core Action                  | Not Run        |                     |

Approval Request

State

Search

Approvals

| State    | Approver             | Comments | Approval for | Created             |
|----------|----------------------|----------|--------------|---------------------|
| Rejected | System Administrator |          | (empty)      | 2026-09-20 06:50:38 |

1 to 1 of 1

Workflow Studio
Network Request
Flow • Automated Network Req.
Network Request
Flow execution • None

EXECUTION DETAILS
Network Request
Test Run - Completed
Open flow
Open context record

Show Action Details
State
Start time

FLOW STATISTICS

Run as: System Administrator   [Open flow logs](#)

**Completed**   2026-09-20 06:50:38   196ms

TRIGGER

Catalog Item Requested

ACTIONS

|     |                                                   |             |                          |                     |       |
|-----|---------------------------------------------------|-------------|--------------------------|---------------------|-------|
| 1   | <b>Get Catalog Variables</b> from Network Request | Core Action | <b>Completed</b>         | 2026-09-20 06:50:38 | 20ms  |
| 2   | <b>Create Record</b>                              | Core Action | <b>Completed</b>         | 2026-09-20 06:50:38 | 4ms   |
| 3   | <b>Send Email</b>                                 |             | <b>Completed</b>         | 2026-09-20 06:50:38 | 24ms  |
| 4   | <b>Ask For Approval</b>                           | Core Action | <b>Completed</b>         | 2026-09-20 06:50:38 | 139ms |
| ▼ 5 | <b>If If Request is Approved</b>                  | Flow Logic  | <b>Evaluated - False</b> | 2026-09-20 06:51:34 | 0ms   |
| 6   | <b>Update Record</b>                              | Core Action | <b>Not Run</b>           |                     |       |
| ▼ 7 | <b>Else</b>                                       | Flow Logic  | <b>Completed</b>         | 2026-09-20 06:51:34 | 4ms   |
| 8   | <b>Update Record</b>                              | Core Action | <b>Completed</b>         | 2026-09-20 06:51:34 | 4ms   |

**Fig-7: Approved and Rejected outcomes, and the Approval Related List on a request record**

## 8. Advantages & Disadvantages

### Advantages

- A single standardized entry point eliminates inconsistent request information.
- Low-code build (no custom scripts) keeps long-term maintenance overhead low.
- End-to-end traceability from submission through approval to final status.
- Automatic notifications and auto-population reduce manual effort and data-entry errors.
- Inherits ServiceNow's native, SLA-backed multi-instance cloud scalability and default security (ACLs) at no extra build cost.
- The same pattern can be reused for other IT service requests with minimal rework.

### Disadvantages

- The approval workflow uses a single “Anyone approves” rule with one approver used during testing, not a multi-level manager/security hierarchy.
- No integration with real network-device provisioning systems (e.g. Cisco DNA Centre, Ansible); approval does not trigger actual device configuration.
- Some connection-specific details still require manual entry by the requester.
- No reporting or analytics dashboard exists yet for tracking request volume or approval turnaround over time.
- The solution depends on ServiceNow platform availability and licensing rather than being a standalone application.
- No formal load/performance testing at scale was conducted.

## 9. Conclusion

This project successfully replaced a manual, fragmented network-request process with a standardized, low-code ServiceNow workflow. Using only native ServiceNow capabilities — a Service Portal catalog item, Catalog Variables, a Catalog UI Policy, a User Information Variable Set, a custom backend table, and a Flow Designer flow — the solution validates and auto-populates request information, creates a traceable backend record automatically, sends notifications, and manages the full approval lifecycle without manual coordination. End-to-end testing confirmed that mandatory-field validation, dynamic form behaviour, auto-population, record creation, email notification, and the Approved/Rejected decision path all function correctly, and a real troubleshooting event (the catalog item's Flow/Process Engine association) was identified and resolved during development. The result meets the project's original objectives: a standardized request form, automatic backend processing, and full traceability from submission to final status.

## 10. Future Scope

The following are realistic next steps identified during brainstorming and prioritization, but were not part of this project's delivered scope:

- Network automation integration (Cisco DNA Centre, Ansible) so an approved request could trigger real device configuration.
- ServiceNow Integration Hub connections to external network-management systems.
- Reporting dashboards (e.g. Power BI or Tableau) for request volume, approval turnaround time, and completion statistics.
- Extending the same catalog item + variable set + UI Policy + Flow Designer + Approval Engine pattern to other IT services, such as device provisioning, access requests, and other infrastructure requests.

## 11. Appendix

### Source Code (if any)

This project is delivered primarily through ServiceNow configuration — Catalog Variables, a Catalog UI Policy, a Variable Set, and a Flow Designer flow — rather than custom scripts. There are no custom Script Includes or Business Rules to include as source code. If required by evaluators, export the Update Set XML for the scoped application and attach it separately, or include configuration screenshots in Section 7.1.

### Dataset Link

Not applicable — the project does not use an external dataset. All data is generated from user-submitted requests through the ServiceNow Service Portal.

### GitHub & Project Demo Link

**GitHub repository:** <https://github.com/irshad56/automated-network-request-management.git>

**Project demo video link(Detailed):** [https://drive.google.com/file/d/1Z6P\\_lSzeWeHzCpK2lcalsE-QAzAkEPw1/view?usp=sharing](https://drive.google.com/file/d/1Z6P_lSzeWeHzCpK2lcalsE-QAzAkEPw1/view?usp=sharing)